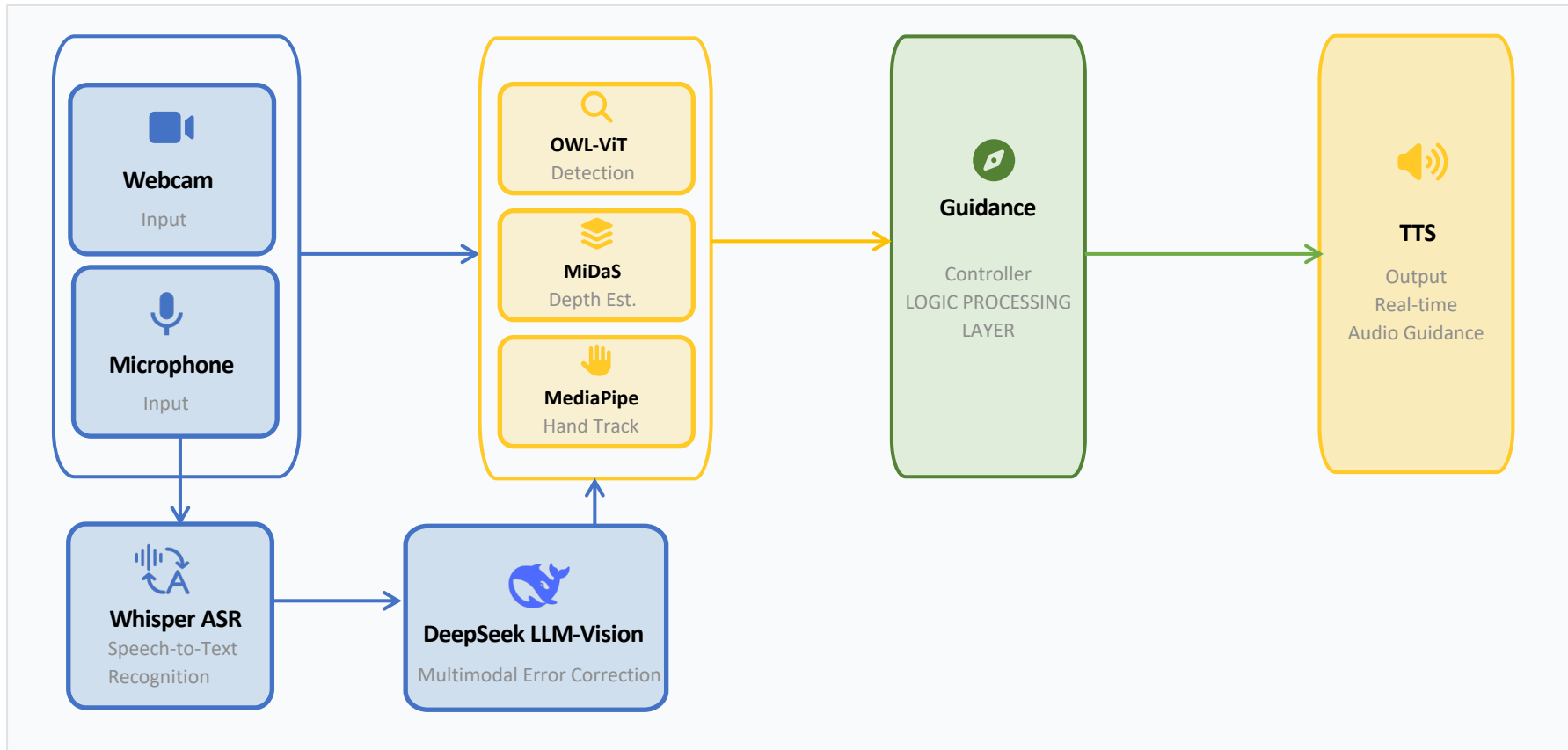


CV Assist: A Real-Time Visual Assistance System for the Visually Impaired

Empowering Independence Through Open-Vocabulary Detection,
Monocular Depth, Hand Tracking & LLM-Vision

Developed by Group 1

Complete Data Flow Architecture



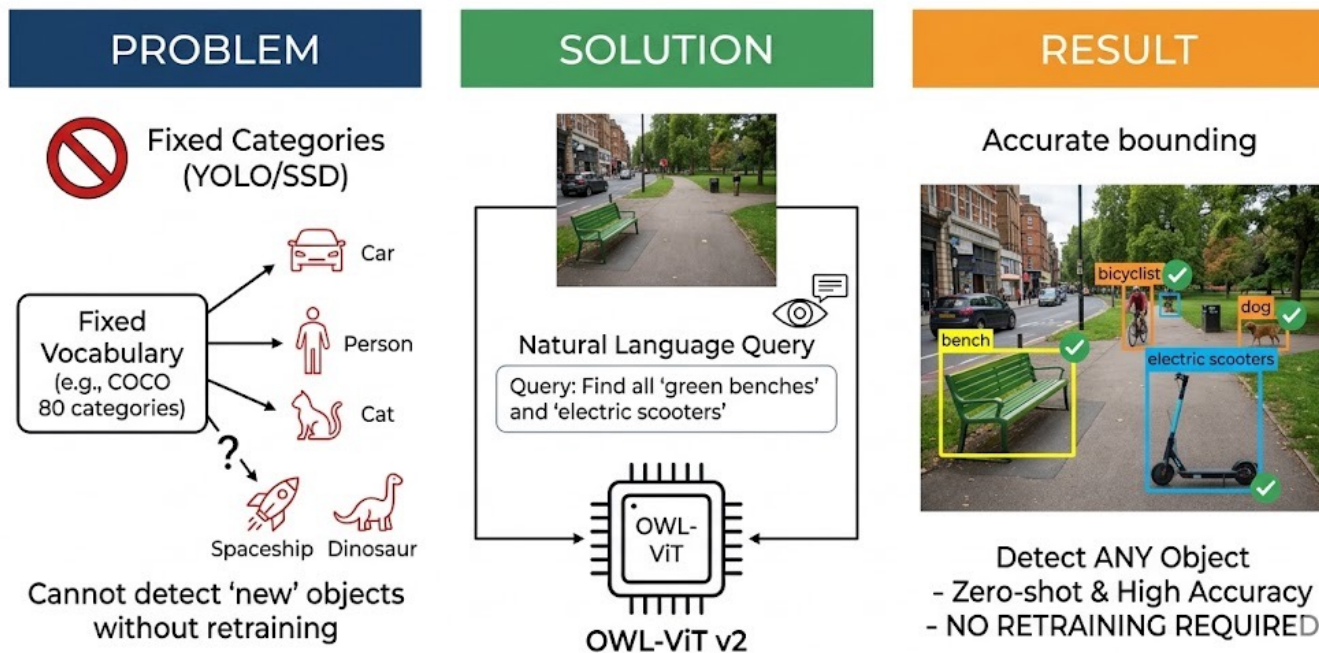
Search for Any Object by Name

Model: OWL-ViT V2

Utilizes the Open-Vocabulary Vision Transformer, a CLIP-based based detector that accepts free-text queries instead of relying on relying on predefined classes.

Key Capability

Open-vocabulary, zero-shot detection from natural language language queries.



Accurate Distance Sensing from a **Single Consumer** Camera

CORE TECHNOLOGY

MiDaS — a lightweight monocular depth estimation model optimized for optimized for real-time inference on standard hardware.

FUNCTION

Dynamically generates **per-pixel relative depth** maps from a single RGB stream, enabling precise distance estimation between the user's hand and the target object without additional sensors.

KEY ADVANTAGE

Eliminates the need for expensive stereo cameras or LiDAR — runs entirely entirely on any standard consumer-grade webcam.



High-Fidelity 3D Hand Landmarks Enable Precise Spatial Alignment

TECHNOLOGY

Google MediaPipe Hands tracks 21 3D hand landmarks in real time, providing robust skeletal tracking from a standard RGB feed.

HAND POSITION

The centroid of key landmarks (wrist + finger bases) is calculated as the definitive "hand location" for spatial offset computation.

GESTURE STATES

OPEN HAND: Reaching / Ready to grasp.

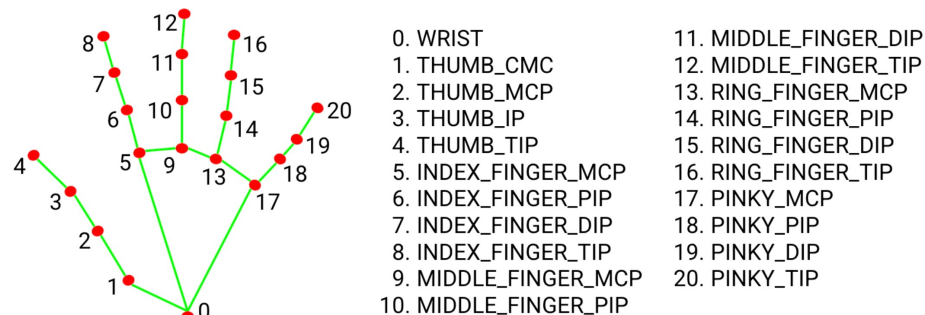
CLOSED FIST: Object grabbed; task complete.

SPATIAL ALIGNMENT

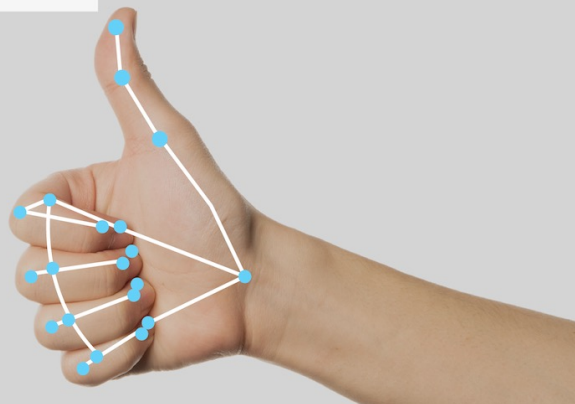
The **X-Y offset** between the hand centroid and the object center drives the directional audio instructions.

PRECISION

Sub-centimeter accuracy enables fine-grained, highly responsive guidance as the hand approaches the target.



Thumbs up 63%



Spatial Guidance and Control Logic

Guidance Controller

The **GuidanceController** is the system's decision-making core, transforming raw detection data into actionable spatial instructions.

Input: Object bbox, depth value, hand landmarks

Process: Spatial relationship calculation

Output: Natural language instructions

Hysteresis-Based Guidance

Prevents oscillation when hand is near alignment boundaries by implementing **directional hysteresis thresholds**.

Example: Horizontal Alignment

Enter "left" zone: $x < -50\text{px}$

Exit "left" zone: $x > -30\text{px}$

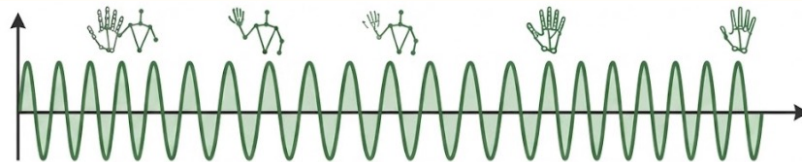
Enter "right" zone: $x > 50\text{px}$

Exit "right" zone: $x < 30\text{px}$

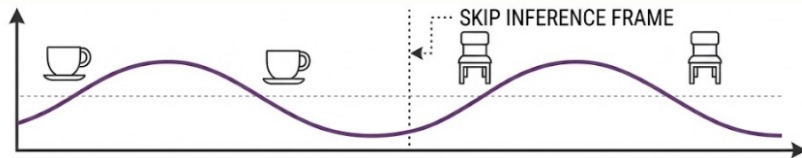
20px deadband prevents rapid instruction switching

DYNAMIC FRAME SKIPPING

HIGH-FREQUENCY (30 HZ): HAND TRACKING (MediaPipe)



LOW-FREQUENCY (5 HZ): OBJECT DETECTION (OWL-ViT/MiDaS)



Intelligently skip LLM inference for object & depth when motion is low, maintaining high perceived smoothness on ordinary devices.

Voice Interaction: ASR and TTS

🎧 ASR: Automatic Speech Recognition

Engine:

OpenAI Whisper (local execution)

Language:

Chinese (zh)

English (en)

🗂️ **Activation:** Press 'v' key to start voice input



🔊 TTS: Text-to-Speech

Real-Time Options

pyttsx3

Basic quality, no setup required

Offline

MiMo (Xiaomi)

High quality, requires API key

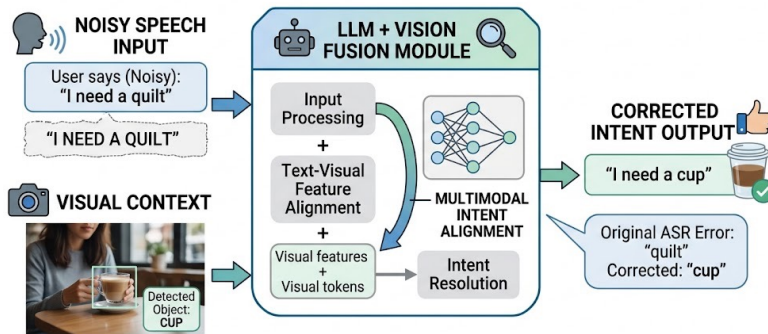
Cloud

Cache Option

Local Cache

Offline Support

High Light: LLM-Assisted Keyword Extraction



Whisper transcribes the command; [DeepSeek-v3](#) uses both to return the most likely object name based on visual context.

The system uses **LLM-Vision multimodal fusion** to correct ASR errors in real time, achieving precise user intent alignment.

Dual-Threshold Logic Delivers Stable, Professional-Grade Audio Instructions

THE PROBLEM: INSTRUCTION JITTER

Naive threshold-based guidance causes rapid oscillation ("Left... Right... Left...") when the hand is near the alignment boundary.

THE SOLUTION: HYSTERESIS ALGORITHM

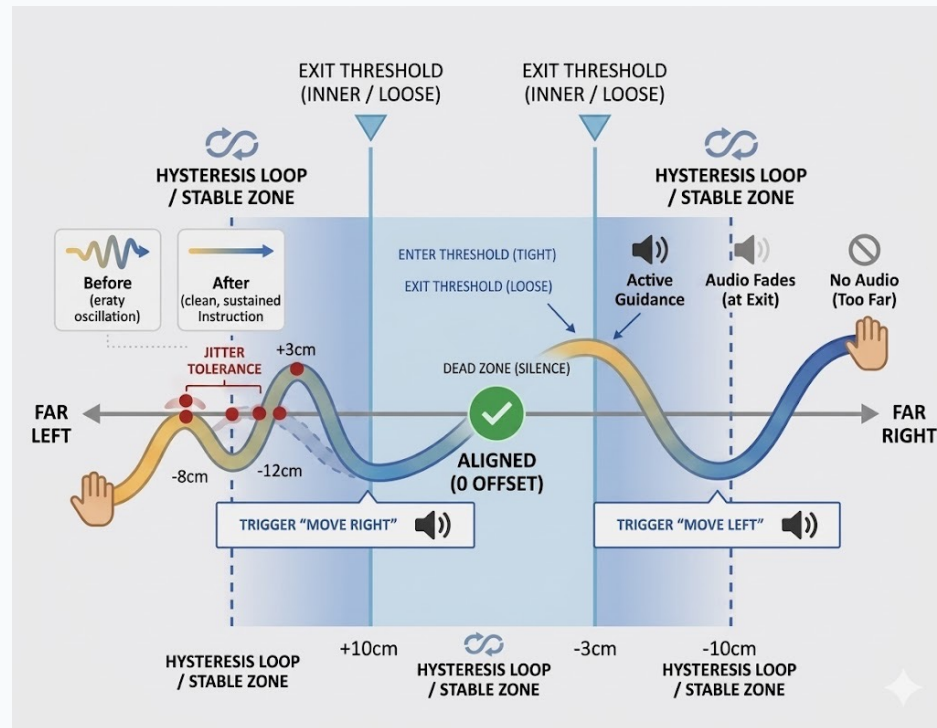
Implements two distinct thresholds for state transitions:

Enter Threshold (Outer): A new directional instruction is triggered only when the offset exceeds a larger boundary.

Exit Threshold (Inner): The instruction is cancelled only when the offset drops below a smaller boundary.

THE RESULT: STABLE GUIDANCE

Creates a stable "dead zone" around alignment. Instructions are issued decisively and cancelled cleanly, ensuring calm and confident spatial audio feedback.



1D Spatial Offset Axis demonstrating dual-threshold state transitions for transitions for jitter-free interaction.

Motivation and Dataset Construction

OWL Fine-tuning for Assistive Object Detection



Motivation

OWL models offer **open-vocabulary detection** capabilities, enabling recognition beyond fixed categories—ideal for **scene-adaptive vocabulary expansion**.

For blind assistance, we prioritize recognizing **high-frequency items** with **safety & convenience value** in daily life.



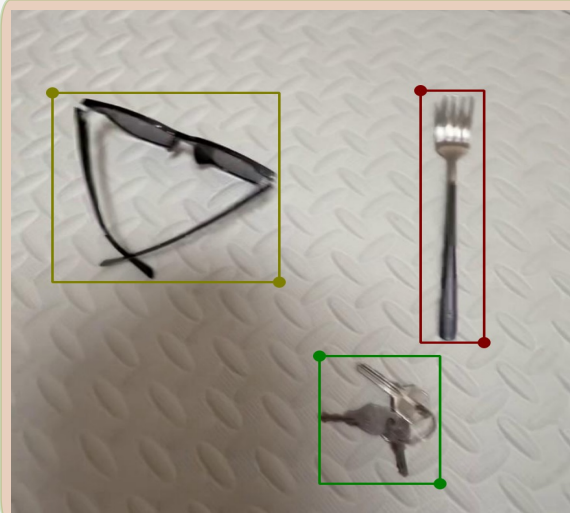
Key
Security



Sunglasses
Daily Use



Stainless Fork
Dining



Data Collection Pipeline

1

Video Recording

Multi-lighting, angles, backgrounds



2

Frame Extraction

Fixed FPS sampling



3

Annotation

Labelme bounding boxes



4

Augmentation & Split

Basic augmentation → train/val



Dataset Statistics

Metric	Value
Total Images	291
Total Annotations	379
Training Set (Images/Boxes)	234 / 305
Validation Set (Images/Boxes)	57 / 74
Number of Classes	3

Fine-tuning Techniques and Results

Technical Approach & Performance Evaluation



Key Techniques



Freeze Backbone

Reduce **overfitting risk** on small datasets while preserving pre-trained visual capabilities.



Contrastive Alignment

Enhance **image-text semantic alignment** through contrastive learning constraints for better query understanding.

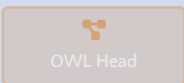
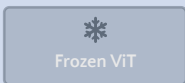
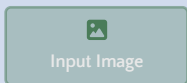


Multi-Loss Optimization

Classification + Box Regression losses jointly optimize for improved localization and classification consistency.



Architecture Overview



● Classification ● Regression



Performance Results (Validation Set)

Model	Status	mAP@0.5	Macro F1	Avg Conf
OWL-ViT	Pre-trained	0.358	0.460	0.309
	Fine-tuned	0.847	0.755	0.432
OWLv2-base	Pre-trained	0.475	0.512	0.779
	Fine-tuned	0.891	0.841	0.823



OWL-ViT mAP Gain

+136%



OWLv2 mAP Gain

+88%



Best Model

OWLv2



Key Finding

Fine-tuning significantly improves detection performance for specialized items. **OWLv2-base** achieves the best overall results with **0.891 mAP** and **0.841 F1**.

System Performance Metrics

Task-Level Metrics

<code>session_id</code>	Session identifier (UUID)
<code>task_id</code>	Incremental task ID
<code>end_reason</code>	Completion cause
<code>target_visible_frames</code>	Stable detection count
<code>first_target_detected_ts</code>	First detection timestamp
<code>report_status</code>	JSON report emitted

Task Lifecycle Timing

<code>s2act</code>	Search-to-Activate
<code>task_duration</code>	Wall-clock time
<code>sup1</code>	Post-suppression processing
<code>tts1</code>	First TTS offset
<code>confirm_window</code>	Pre-activation duration

Resource Monitoring

<code>cpu</code>	Process CPU utilization
<code>mem</code>	Resident Set Size (RSS)
<code>gpu</code>	CUDA allocated memory
<code>net↑</code>	Network upload throughput
<code>net↓</code>	Network download throughput

Frame Processing Pipeline

<code>det1</code>	Object detection (OWL-ViT)
<code>hand1</code>	Hand detection/tracking
<code>d1</code>	Depth estimation (MiDaS)
<code>g1</code>	Guidance calculation
<code>proc1</code>	Total algorithm processing
<code>e2e1</code>	End-to-end frame loop

FPS & Latency Stats

<code>fps_avg</code>	Average processing FPS
<code>e2e_avg</code>	Average end-to-end FPS
<code>p50</code>	Processing time median
<code>p95</code>	95th percentile
<code>jitter</code>	Standard deviation
<code>det_run</code>	Detection execution rate
<code>depth_run</code>	Depth estimation rate

Voice Metrics

<code>poe</code>	Poe vision parsing
<code>asr_latency</code>	Speech recognition time
<code>record_time</code>	Audio recording duration
<code>first_tts_latency</code>	First TTS response
<code>cam_fail</code>	Camera read failure count

Real-time Process Supervision Logs

</> cmd

[2026-04-14 10:36:39] [core.system] [INFO] FPS 统计
[帧 200] | 处理: 当前=91.4, 平均=34.4, 最小=3.9, 最大=100.7 | 端到端: 当前=64.8, 平均=29.4, 最小=3.8, 最大=71.4

[2026-04-13 23:06:26] [core.system] [INFO] [task] |
task_0005 | a scissors | running(进行中) | 35.3s |
s2act=4.3s | poe=1.8s | det1=78.0ms | hand1=16.7ms |
d1=61.5ms | g1=0.0ms | proc1=156.3ms |
e2e1=189.0ms | sup1=147.4ms | tts1=+29.1s |
det_run=33.5% | depth_run=33.5% | fps_avg=16.2 |
e2e_avg=12.9 | p50=16.9ms | p95=153.7ms |
jitter=63.2ms | cam_fail=0.0 | gpu=1632.6MB

<> task_metrics_2026-04-14_task_01.json

分区	中文说明	英文键	值
会话信息	会话 ID	session_id	cf0b1c16be9d4bbe9402c37913ad
会话信息	报告生成时间	generated_at	2026-04-13 22:36:54.587 HKT
...	...	...	...
任务信息	任务 ID	task_id	task_0001
任务信息	目标主体	target_query	a phone
任务信息	任务状态	task_state	finished (已结束)
任务信息	任务耗时秒	task_elapsed_sec	39.718
任务信息	结束原因	end_reason	success (成功完成)
...	...	...	...
运行摘要	总帧数	total_frames	530
运行摘要	首次生成引导时间戳	first_guidance_ts	2026-04-13 22:36:24.658 HKT
运行摘要	GPU 显存占用 MB	gpu_memory_mb	1632.9
...	...	...	...
语音摘要	语音总耗时秒	voice_total_time_sec	2.818
语音摘要	Poe 视觉解析耗时秒	llm_poe_sec	2.174
语音摘要	语音原文	raw_text	呃, 我想要拿我的手机
...	...	...	...
时延摘要	平均处理耗时秒	process_time_avg_sec	0.057
时延摘要	平均端到端耗时秒	e2e_loop_time_avg_sec	0.075
...	...	...	...

Empowering Independence Through Intelligent Assistance

Future work will expand gesture vocabulary and optimize deployment for edge devices. By integrating haptic feedback, we aim to evolve the system into a multi-sensory ecosystem for more intuitive human-spatial interaction.



Computer Vision

Open-vocabulary detection with zero-shot capability



AI Integration

LLM-enhanced voice recognition with visual context



Human-Centered

Designed for real users with real accessibility needs